

Docker Task -3

Task Description:

Create a custom docker image for nginx and deploy it using docker compose, where the volume bind mount should be at /var/opt/nginx location. Push the created custom docker image to your docker-hub.

Step 1 - Installed docker and docker-compose

sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-20-53:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-01-27 05:40:59 UTC; 14s ago
     TriggeredBy: ● docker.socket
   Main PID: 2350 (dockerd)
      Tasks: 9
     Memory: 115.8M (peak: 116.3M)
        CPU: 456ms
      CGroup: /system.slice/docker.service
              └─2350 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Jan 27 05:40:58 ip-172-31-20-53 dockerd[2350]: time="2026-01-27T05:40:58.644459660Z" level=info msg="Restoring containers: start."
Jan 27 05:40:58 ip-172-31-20-53 dockerd[2350]: time="2026-01-27T05:40:58.739094537Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Jan 27 05:40:58 ip-172-31-20-53 dockerd[2350]: time="2026-01-27T05:40:58.747208532Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Jan 27 05:40:59 ip-172-31-20-53 dockerd[2350]: time="2026-01-27T05:40:59.168439126Z" level=info msg="Loading containers: done."
Jan 27 05:40:59 ip-172-31-20-53 dockerd[2350]: time="2026-01-27T05:40:59.252404080Z" level=info msg="Docker daemon" commit=9c62384 containerd-snapshotter=true storage-driver=overlayfs vers
Jan 27 05:40:59 ip-172-31-20-53 dockerd[2350]: time="2026-01-27T05:40:59.252521686Z" level=info msg="Initializing buildkit"
Jan 27 05:40:59 ip-172-31-20-53 dockerd[2350]: time="2026-01-27T05:40:59.548544989Z" level=info msg="Completed buildkit initialization"
Jan 27 05:40:59 ip-172-31-20-53 dockerd[2350]: time="2026-01-27T05:40:59.555154963Z" level=info msg="Daemon has completed initialization"
Jan 27 05:40:59 ip-172-31-20-53 systemd[1]: Started docker.service - Docker Application Container Engine.
Jan 27 05:40:59 ip-172-31-20-53 dockerd[2350]: time="2026-01-27T05:40:59.555319538Z" level=info msg="API listen on /run/docker.sock"
```

Step 2 - Created a working directory nginx-custom and created html file, Dockerfile and docker-compose.yml

Dockerfile :

FROM nginx:alpine

Remove default nginx content

RUN rm -rf /usr/share/nginx/html/*

Copy custom content

COPY html/ /usr/share/nginx/html/

Expose port

EXPOSE 80

docker-compose.yml :

version: "3.9"

services:

nginx:

image: govindcloud01/custom-nginx:1.0

container_name: custom-nginx

ports:

- "8080:80"

volumes:

- /var/opt/nginx:/usr/share/nginx/html

Step 3 : Prepare host bind mount directory

```
sudo mkdir -p /var/opt/nginx
```

```
sudo cp html/index.html /var/opt/nginx/
```

```
sudo chmod -R 755 /var/opt/nginx
```

Step 4 - Build and push the image to docker hub

```
docker login -u govindcloud01
```

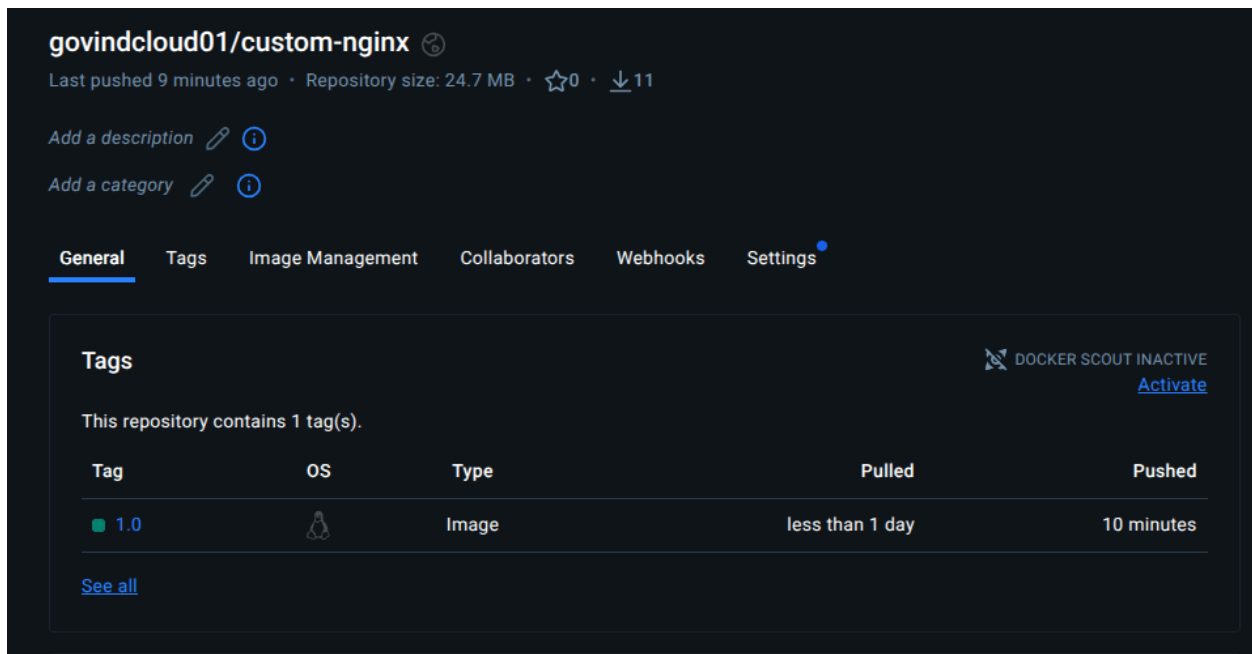
```
docker build -t your_dockerhub_username/custom-nginx:1.0 .
```

```
docker images
```

```
docker push your_dockerhub_username/custom-nginx:1.0
```

```
docker compose up -d
```

hub.docker.com :



The screenshot shows the Docker Hub interface for the repository **govindcloud01/custom-nginx**. The repository was last pushed 9 minutes ago and has a size of 24.7 MB. It has 0 stars and 11 downloads. Below the repository name, there are links to "Add a description" and "Add a category". The "General" tab is selected, showing a list of tags. The "Tags" section indicates that the repository contains 1 tag(s). A table lists the tags, with one tag named "1.0" shown. The table columns are Tag, OS, Type, Pulled, and Pushed. The tag "1.0" is an Image, pulled less than 1 day ago, and pushed 10 minutes ago. There is a "See all" link below the table. In the top right corner, there is a "DOCKER SCOUT INACTIVE" notification with an "Activate" link.

govindcloud01/custom-nginx ⓘ

Last pushed 9 minutes ago · Repository size: 24.7 MB · ☆0 · ↓11

[Add a description](#) ⓘ ⓘ

[Add a category](#) ⓘ ⓘ

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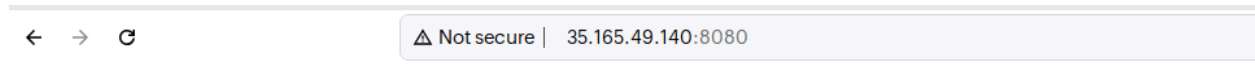
Tags DOCKER SCOUT INACTIVE [Activate](#)

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
1.0		Image	less than 1 day	10 minutes

[See all](#)

html page is accessible in browser :



NGINX running from Custom Docker Image by GOVIND ðŸš€

Volume mounted at /var/opt/nginx